INNOVATION PROPOSAL · BAOR MASTER

An integrated matching system for the BAOR Master.

Thesis allocation · Early-career matching · Career-services analytics

Business Analytics & Operations Research · TiSEM, Tilburg University · 2026

Built on the Rookies matching engine

Three problems. One mathematics.

The BAOR master runs three allocation processes that are, structurally, the same problem: matching thesis students to topics and supervisors, matching graduates to first jobs, and guiding students toward the careers that fit them. Each is a two-sided matching market. Each is run by hand today. Mechanism design solves all three with one engine.

How the BAOR master allocates today.

Three processes, each run manually, none informed by the others.

Thesis allocation

[INSERT CURRENT ALLOCATION METHOD] …coordinators reconcile by hand. No shared view of fit, fairness, or capacity.

Job placement

Students apply by guesswork; employers screen by keyword; no one computes fit. The allocation is left to chance, and the program never sees where the cohort lands.

Career guidance

Advising rests on intuition and last year's anecdotes — not on what this cohort's data, and the live labour market, actually show.

Stable matching is solved mathematics.

The department already teaches it. This proposal puts it to work inside the department.

1962

Gale & Shapley

Stable matching.

2012

Roth & Shapley

Nobel Prize — market design as engineering.

2025

NRMP

37,667 doctors placed in a single match.

The same framework that places medical residents, allocates school seats, and clears kidney-exchange networks applies directly to thesis allocation and early-career hiring. The expertise to deploy it already sits in Econometrics & OR.

One engine. One data layer. Three applications.

ONE SYSTEM

Matching engine

Quantitative scoring · optimisation

Reasoning-layer AI · rationale per pair

Stable matching · Gale–Shapley / Roth

Data layer

Longitudinal cohort data

Every match teaches the next

THREE APPLICATIONS

Thesis matching

Students ↔ topics ↔ coordinators

Job matching

Students ↔ early-career roles

Career-services portal

Dashboard · analytics · guidance

Thesis allocation as a matching market.

A three-sided problem: students, topics, and the coordinators who own them.

Each student–topic pair receives a compatibility score from the matching engine, blending methodological fit, prior coursework, and stated interest. The algorithm then returns the assignment that maximizes total compatibility across the cohort, subject to each coordinator's supervision capacity and expertise. Individual fit and the strength of the cohort-wide allocation are produced together, the best overall set of theses the programme can run this year.

Fair & stable

No back-room reshuffles. Every allocation is defensible.

Less manual work

Coordinators set preferences and capacity once — not weeks of email.

Better-fit theses

Matches reflect methodological fit, not who emailed first.

Auditable

Every assignment carries a reason. Transparent by construction.

Early-career matching for BAOR graduates.

The same engine, pointed at the first job instead of the thesis.

A BAOR student builds one profile: courses, projects, skills, ambitions. The matching engine scores fit against live roles: a quantitative layer over structured features, and a reasoning layer for the qualitative signal that mathematics cannot capture. The student receives a short list of ranked, explained matches: five roles worth applying to, not a hundred to spray.

Quantitative scoring

Skills, courses, and project signal scored against role requirements. Deterministic and fast.

Reasoning layer

Reads projects and self-descriptions for the qualitative fit, with a rationale for every score.

Explainable by design

Per-pair transparency — aligned with the EU AI Act for high-risk recruitment systems.

A career-services portal for BAOR.

The Career Officer dashboard turns cohort data into decisions. Four tools.

Match Analyzer

Pick a student and a target role. See the match score, where the gaps are, and what to raise in the 1-on-1.

Market Pulse

What employers are searching for at Tilburg right now: trending skills, sector demand, hot roles. Refreshed weekly.

Career Path Explorer

Type a role. See the skills employers ask for, the backgrounds of students who placed, and what lifts an application.

Engagement & At-risk

Who's slipping, who's progressing, and where the team should focus outreach this month.

08 · LIVE DEMO

Built, not (only) pitched.

demo version

Market Pulse

What employers are searching for at TiU right now — refreshed weekly.

POSTINGS THIS MONTH	DISTINCT SKILLS TRACKED	ACTIVE EMPLOYERS	AVG. APPLICATIONS / POSTING
142	487	68	14.2
↑ 23% vs. April	↑ 31 new this month	↑ 9 new entrants	↓ 1.4 vs. April

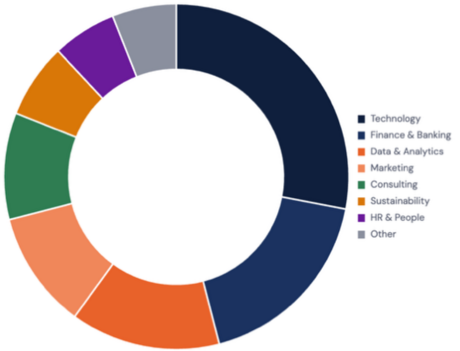
TRENDING SKILLS · MAY 2026

vs. previous month

01	Python	47	↑ 12
02	SQL	41	↑ 8
03	Excel	38	↑ 3
04	Power BI	32	↑ 15
05	Financial modelling	28	↑ 5
06	Dutch (B2+)	26	↓ 2
07	R	24	↑ 9
08	Tableau	22	↑ 4
09	Stata	19	↑ 1
10	PowerPoint	18	↓ 3

SECTOR DEMAND

% of postings



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Engagement & At-risk students

Who's slipping, who's progressing, and where to focus your team's outreach this week.

AT-RISK STUDENTS · TISEM MASTER'S

Final-year cohort · 8 flagged this week

STUDENT	ISSUE	SUGGESTED ACTION	
M Maaike de Boer	Final-year MSc · No applications · Last login 35 days ago	Send 1-on-1 invite	Open →
J Jasper de Vries	Final-year BSc · Profile only 30% complete · No CV uploaded	Profile completion prompt	Open →
N Nadia van Beek	MSc · 6 months to grad · No employer messages received	Visibility / discoverability check	Open →
L Lukas Maas	Pre-master · No skills listed on profile	Skills inventory session	Open →
S Sara Dekker	Final-year · 2 applications submitted · 0 responses	CV review + tailoring session	Open →
T Tom Bakker	MSc · Logged in once in past 60 days	Re-engagement email	Open →
E Eva Visser	Final-year BSc · No location preferences set · Likely missing matches	Career-direction session	Open →
D Daan Hendriks	MSc · Profile complete but no applications in 4 weeks	Application coaching	Open →

COHORT ENGAGEMENT

TiSEM MSc · year 2 · 412 students

Active profiles	287 / 412 · 70%
Profile complete (≥75%)	198 / 412 · 48%
Submitted ≥1 application	127 / 412 · 31%
Placed (offer accepted)	43 / 412 · 10%

Heads-up. Application rate is 9 points behind same week last year. Consider a targeted nudge to the 88 students with complete profiles but zero applications.

What the BAOR student gets.

One profile, built once, working across the whole student lifecycle.

Ranked, explained thesis options	Topics that fit the student's methods and interests: with the reason for each, not a first-come scramble.
Ranked, explained job matches	A short list of roles worth applying to, each with a rationale. No spray-applying.
A clear view of skills gaps	What the market asks for, what the student has, and the concrete steps that close the distance.
Continuity from thesis to first job	The same profile and the same engine carry the student from topic selection through placement.

Every match teaches the system.

Each allocation and each placement feeds a longitudinal record: which student–topic matches produced strong theses, which placements held, what skills the market demanded and when. That record improves every future match, and it is, in itself, a research asset.

Better matches

Outcome data tunes the engine. The next cohort's allocation is sharper than this one's.

Program insight

Skills-gap and placement signal feed curriculum and advising decisions with evidence.

Research output

A dataset on revealed-preference matching under information asymmetry — publishable, and on the department's own line of work.

Innovation?

A first in NL higher education	The first application of stable-matching mechanism design to an integrated academic-and-career student lifecycle.
Explainable by construction	Every match carries a rationale, aligned with the EU AI Act's transparency rules for high-risk systems.
A research dataset	Generates publishable data on matching under information asymmetry, on the department's existing research line.
Replicable across TiSEM	Built once for BAOR, the architecture extends to every master's programme in the school.
Concrete operational wins	Coordinator hours saved, fairer thesis allocation, and measurable placement outcomes from year one.

The first programme with a matched student lifecycle.

From the first thesis topic to the first job, every BAOR student allocated by the same mathematics the department teaches. Better outcomes for students, fewer manual hours for coordinators, and a dataset that turns the programme's own operations into research.